

# The Measurement Problem: 0/0 of Quantum Measurement

Michael Grafiel S Puno

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The Removable Singularity Framework

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## Abstract

We show that the quantum measurement problem IS a 0/0 removable singularity: before measurement, the system is in superposition (0/0); after measurement, it is definite. The Born rule ( $P = |a|^2$ ) converts certainty to probability. Decoherence explains why we don't see superpositions. The many-worlds interpretation eliminates collapse but creates  $10^{30}$  branches. The quantum Zeno effect shows measurement CREATES reality. Wheeler's delayed choice shows the present affects the past. This connects consciousness, entanglement, holography, the arrow of time, and black holes.

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## 1. Superposition (Quantum 0/0)

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Before measurement, a quantum system exists in BOTH states simultaneously. This is the 0/0: the system is in multiple states at once. The Schrodinger's cat thought experiment illustrates this: before opening the box, the cat is BOTH alive and dead.

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## 2. Born Rule

$$P(0) = |a|^2, P(1) = |b|^2$$

Measurement outcomes are PROBABILISTIC. The Born rule converts certainty (superposition) to probability. This is the 0/0: probability emerges from certainty.

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## 3. Decoherence

The environment "measures" the system, causing off-diagonal elements of the density matrix to decay. This explains WHY we don't see superpositions in everyday life. Decoherence is NOT collapse -- it converts quantum superposition to classical probabilities.

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## 4. Many-Worlds Interpretation

No collapse! Instead, branching: each measurement creates TWO branches. After 100 measurements:  $10^{30}$  branches. ALL branches are equally real. The 0/0 is resolved by splitting reality.

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## 5. Quantum Zeno Effect

Frequent measurement prevents evolution! If you measure often enough, the system NEVER changes. Measurement CREATES reality: the act of observing defines what exists. This is the deepest 0/0.

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## 6. Delayed Choice Experiment

Wheeler (1978): we can decide to measure particle/wave AFTER the particle has passed through slits. The present affects the past. The act of measurement CREATES the history.

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## 7. Connections

| System        | Prior Chapter       | Connection      |
|---------------|---------------------|-----------------|
| Consciousness | Ch.34 Consciousness | Observer effect |
| Entanglement  | Ch.33 Entanglement  | Quantum info    |
| Holographic   | Ch.47 Holographic   | Info boundary   |
| Arrow of time | Ch.48 Arrow of time | Decoherence     |
| Black holes   | Ch.32 Black holes   | Info paradox    |
| RMT           | Ch.44 RMT           | Spectral stats  |

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## 8. Conclusion

The measurement problem IS a 0/0: superposition -> measurement -> collapse. The Born rule converts certainty to probability. Decoherence explains classicality. Many-worlds eliminates collapse but creates branching. The quantum Zeno effect shows measurement CREATES reality. Delayed choice shows the present affects the past. This is the deepest 0/0 in physics, connecting consciousness, entanglement, holography, the arrow of time, and black holes.

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## References

- [1] Born (1926), statistical interpretation of quantum mechanics
- [2] Zeh (1970), decoherence interpretation
- [3] Everett (1957), many-worlds interpretation
- [4] Wheeler (1978), delayed choice experiment
- [5] Misra & Sudarshan (1977), quantum Zeno effect
- [6] Puno, "The Removable Singularity" (2026)